UNIVERSITY OF MUMBAI • BE CSE (DATA SCIENCE)

GeoTrack

Real-Time Geospatial Telemetry & Trajectory Analysis System

Subject: Wireless & Mobile Comm.
Semester: BE Semester 3
Focus: GNSS, A-GPS & Signal Filtering
Datum: WGS-84 / Web Mercator

1. Project Overview in Brief

GeoTrack is a high-precision, field-ready geospatial telemetry system engineered to interface directly with device hardware radios (satellite GNSS, 4G/5G cellular towers, and Wi-Fi networks). It captures real-time position streams, filters out urban reflection noise using an adaptive **Kalman Filter**, and delivers an interactive mission control dashboard displaying ground speed, heading, altitude, and trajectory logs.

> **Core Engineering Problem Solved:** Raw satellite location data often jumps wildly (by tens of meters) due to atmospheric delays and building reflections. GeoTrack turns noisy, scattered radio measurements into a smooth, accurate, vehicle-grade path in real time.

2. Wireless & Mobile Communication (WMC) Concepts

📡 A. Satellite Trilateration: The "Sphere Intersection" Concept

GPS satellites in medium Earth orbit (20,200 km altitude) continuously broadcast radio signals containing atomic clock timestamps. Your device does **not** send signals back to space; it only listens (downlink only).

Because radio waves travel at the speed of light (**300,000 km/s**), your receiver measures the microsecond delay to calculate its distance from each satellite:

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Distance = Speed of Light (c) × Time Delay (Δt)
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- **1 Satellite:** Places you somewhere on the surface of a giant sphere in space.
- **2 Satellites:** Narrows your location to the circular intersection of two spheres.
- **3 Satellites:** Intersects into two discrete points on Earth.
- **4th Satellite (Crucial!):** Resolves the clock bias between your phone's cheap quartz clock and the satellite's atomic clock, fixing your exact **Latitude, Longitude, and MSL Altitude**.

📶 B. Carrier Frequencies: GPS L1, Dual-Band L5, and NavIC

Satellites transmit across dedicated microwave bands:

- **GPS L1 Band (1575.42 MHz):** The legacy civilian signal. Good coverage, but susceptible to signal reflections off glass buildings.
- **Modern L5 Band (1176.45 MHz):** High-power, wide-bandwidth signal with advanced signal-to-noise ratio (SNR) that filters out bounced waves in urban areas.
- **NavIC (IRNSS):** India’s independent regional navigation satellite system developed by ISRO, operating on L5 and S-band frequencies, tailored for the Indian subcontinent.

🌆 C. Multipath Interference ("Urban Canyons")

When tracking in urban areas surrounded by tall concrete and glass buildings, direct line-of-sight to satellites is obstructed. The radio wave bounces off glass facades and pavements before entering your antenna.

Impact: Because the reflected signal took a longer physical route, the calculated distance is artificially enlarged, creating "*ghost jumps*" (teleporting across buildings).

🚀 D. Assisted GPS (A-GPS) & Cellular / Wi-Fi Positioning (WPS)

Direct satellite signals are faint (roughly -130 dBm). A cold start without assistance can take 1 to 5 minutes to download satellite orbit tables (almanac and ephemeris).

- **A-GPS:** The device connects to 4G/5G cell towers to download orbital ephemeris data in under 2 seconds over the cellular data channel.
- **Wi-Fi Positioning (WPS):** Indoors, where satellite microwaves are blocked by concrete slabs, the phone reads the unique MAC addresses (BSSIDs) of nearby routers and looks them up in global geolocation databases to pinpoint location.

⚠️ Key Viva Concept: Why Laptops Show "Nagpur" When You Are in "Panvel"

Laptops and desktop PCs do **NOT** have GNSS satellite chips. When the browser requests location on a PC, it falls back to **IP Geolocation**. Indian broadband providers route traffic through central regional data centers and NOCs (e.g. Nagpur or Mumbai hubs). The IP lookup returns the location of your ISP's server, with a wide uncertainty radius of **±50,000 meters (50 km)**.

Solution: Opening GeoTrack on a smartphone engages real dual-frequency GNSS hardware, delivering true **±3 to 15 meter** accuracy in Panvel.

3. Noise Filtering: The Kalman Filter Explained Simply

Raw GPS coordinates bounce erratically due to atmospheric jitter. GeoTrack eliminates this using a two-stage filter:

🧠 The Intuitive "Two Friends" Kalman Analogy

Imagine you are walking down a road with your eyes closed:

- **Friend 1 (The Physics Predictor):** Knows your walking velocity. He predicts: "*Shivam was at meter 10, walking at 1 m/s, so after 1 second he must be near meter 11.*"
- **Friend 2 (The GPS Sensor):** Opens his eyes with blurry vision and says: "*I see him at meter 13, but my accuracy is ±4 meters.*"
- **The Kalman Algorithm:** Calculates the mathematical balance between the two. When GPS accuracy is sharp (low HDOP), it relies heavily on the GPS reading. When GPS signal is degraded or fluctuating, it relies on the physical prediction, producing a smooth trajectory without jitter.

🛡️ Velocity-Gated Jump Filter & Stationary Deadband

- **Teleport Rejection:** Calculates implied velocity between consecutive points. If the implied speed exceeds **300 km/h** and falls outside the uncertainty circle, it is instantly discarded as a multipath artifact.
- **Stationary Deadband:** When stationary at a traffic signal or desk, satellite noise makes coordinates drift. Any movement under **1.5 meters** (or under 2% of the accuracy circle) is flagged as stationary noise.

4. System Architecture & Live Telemetry Metrics

Telemetry Metric	Wireless & Physical Meaning	Field Use Case
Coordinates (Lat/Lng)	Angular position on WGS-84 ellipsoidal Earth model	Pinpoint location mapping
Accuracy (HDOP)	Horizontal Dilution of Precision (satellite geometry quality)	Confidence radius (in meters) for position fix
Instant Ground Speed	Haversine differential over delta time (km/h)	Vehicle and pedestrian transit analysis
Bearing / Heading	Forward azimuth relative to True North (0°–360°)	Compass heading & directional orientation
MSL Altitude	Height above Mean Sea Level calculated via 4+ satellites	Topographical and elevation profiling
Mission Clock	Continuous elapsed mission time since first lock	Time-distance integral for route profiling

5. Slide-by-Slide Speaking Script for Presentation

Use these exact spoken points during your 7-slide committee evaluation:

Slide 1 — Title & Introduction

"Good morning respected evaluators. Today we present GeoTrack, an intelligent real-time geospatial telemetry and trajectory tracking platform designed around the principles of Wireless and Mobile Communication."

Slide 2 — The Engineering Problem

"Standard civilian GPS suffers from satellite clock drift, atmospheric ionospheric refraction, and urban multipath reflections. Our objective is to capture raw radio streams and apply intelligent filtering to deliver reliable, jitter-free positioning."

Slide 3 — Wireless Positioning Architecture

"Our system receives signals from the space segment (GNSS L1/L5 microwave frequencies), aided by the terrestrial cellular network (A-GPS) and local Wi-Fi BSSID beacons for indoor/rapid time-to-first-fix."

Slide 4 — Noise Filtering & Kalman Smoothing

"Rather than accepting raw, erratic coordinates, GeoTrack runs an adaptive 1D Kalman Filter combined with a velocity-gate jump filter. This prevents false jumps and stationary drift without introducing perceptible latency."

Slide 5 — Real-time Telemetry Dashboard

"Our telemetry engine computes instant ground velocity, heading angles relative to True North, and Mean Sea Level elevation, rendering the path smoothly on an interactive WGS-84 vector map."

Slide 6 — Network Tradeoffs & Classroom Demonstration

"We diagnosed the fundamental distinction between true smartphone hardware GNSS (3–15 meter precision) and desktop IP-based fallback (50 km ISP radius). For classroom evaluations, we built an integrated deterministic trajectory simulator."

Slide 7 — Conclusion & Future Scope

"GeoTrack demonstrates the practical application of wireless physical-layer propagation theory and real-time algorithmic smoothing. We welcome your questions."

6. Anticipated Viva / Examiner Questions & Answers

Q1: Why are four satellites required instead of three for GPS?

> Three satellites provide three spheres that intersect at two points, one of which can be discarded as unrealistic. However, receiver quartz clocks have slight time inaccuracies. Because radio waves travel at 300,000 km/s, even a 1-microsecond clock error causes a 300-meter position error. The 4th satellite solves for this receiver clock bias (Δt), locking 3D coordinates accurately.

Q2: What is HDOP and what does it indicate?

> HDOP stands for *Horizontal Dilution of Precision*. It is a mathematical factor representing satellite geometry. When satellites are spread widely across the sky, HDOP is low (< 2.0) and precision is high. If all visible satellites are clustered in one tight corner, HDOP is high (> 5.0) and horizontal accuracy degrades significantly.

Q3: Why doesn't GPS work inside underground tunnels or metro stations?

> GNSS satellites broadcast in the microwave spectrum (1.1 to 1.5 GHz) with transmitter power of only 25–50 watts across a 20,000 km distance. Concrete, steel reinforcements, and earth completely attenuate and absorb these low-power microwave signals, causing total loss of Signal-to-Noise Ratio (SNR).

Q4: How does your Kalman Filter differ from a simple moving average?

> A simple moving average introduces lag/delay and treats all points equally. The Kalman Filter dynamically weights measurements based on the incoming sensor's HDOP accuracy: if a measurement has high uncertainty, it relies on physical motion prediction; if accuracy is sharp, it favors the sensor reading, maintaining zero tracking lag.
